

Software Requirements Specification (SRS)

Project: Zura Touch – Prize Wheel

Version: 0.1

Date:14/07/2026

1. Project Overview

Objective

The objective of this project is to develop a native Android application for a vending machine that offers an interactive Spin-to-Win experience.

2. User Experience (UX)

Instead of selecting a specific product, users purchase a spin on a prize wheel. The application determines a winning category, communicates with the vending machine and reveals the prize only after the machine confirms that the product has been successfully dispensed.

User Flow

Idle Screen

↓

User purchases a spin

↓

Payment confirmation

↓

Prize category selected

↓

Identify the corresponding shelf and front slot

↓

Command the vending machine to dispense

↓

Machine confirms successful delivery

↓

Determine the prize using the shelf and slot mapping

↓

Reveal the prize

↓

Update stock

↓

Return to Idle Screen

3. Functionalities

Main Screen

- Animated Prize Wheel.
- Company branding.
- Mystery Box price.
- Idle animation.

Prize Wheel

- Displays prize categories instead of individual products.
- Updates dynamically according to product availability.
- Smooth spinning animation.
- Premium prize reveal animation.

Mystery Box Logic

- User purchases a spin.
- Application determines the winning category.
- Application identifies the corresponding shelf and available front slot.
- Machine dispenses the product.
- Prize is revealed after successful delivery confirmation.

Stock Management

- Internal stock tracking.
- Automatic stock update after every successful sale.
- Automatic update of available products.
- Categories with no available products should no longer appear on the wheel.

Error Handling

- Payment errors.
- Dispensing errors.

- Friendly user messages.
- Automatic return to idle mode.

4. Technical Notes

The application should be developed as a native Android application.

Suggested technologies:

- Kotlin
- Jetpack Compose

The application will communicate with the vending machine through the manufacturer's UART communication protocol.

The software architecture should be modular and scalable.

The application should support future cloud integration.

5. Future Improvements

Possible future features include:

- User accounts
- Mobile application
- QR Code login
- Loyalty program
- Cloud synchronization
- Analytics dashboard
- Remote configuration
- Promotional campaigns
- Seasonal themes
- AI-based recommendations
- Multi-language support
- Digital coupons

These features are outside the scope of the first version.

Appendix

The following documents should be provided together with this specification:

- UART Communication Protocol
- Vending Machine Controller (VMC) Communication Protocol